Chapter 1

Example problem: Unsteady flow in a 2D channel, driven by an applied traction

In this example we consider a variation of the unsteady 2D channel flow problem considered elsewhere. In the previous example the flow was driven by the imposed wall motion. Here we shall consider the case in which the flow is driven by an applied traction $\mathbf{t}^{[prescribed]}$ which balances the fluid stress so that

$$t_i^{[prescribed]} = \tau_{ij}\, n_j = \left(-p\, \delta_{ij} + \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right)\right) n_j \qquad (1)$$

along the upper, horizontal boundary of the channel. Here n_j is the outward unit normal, δ_{ij} is the Kronecker delta and τ_{ij} the stress tensor. oomph-lib provides traction elements that can be applied along a domain boundary to (weakly) impose the above boundary condition. The traction elements are used in the same way as flux-elements in the Poisson and unsteady heat examples. The section Comments and Exercises at the end of this documents provides more detail on the underlying theory and its implementation in oomph-lib.

1.1 The example problem

We consider the unsteady finite-Reynolds-number flow in a 2D channel that is driven by an applied traction along its upper boundary.

Unsteady flow in a 2D channel, driven by an applied traction.

Here is a sketch of the problem:

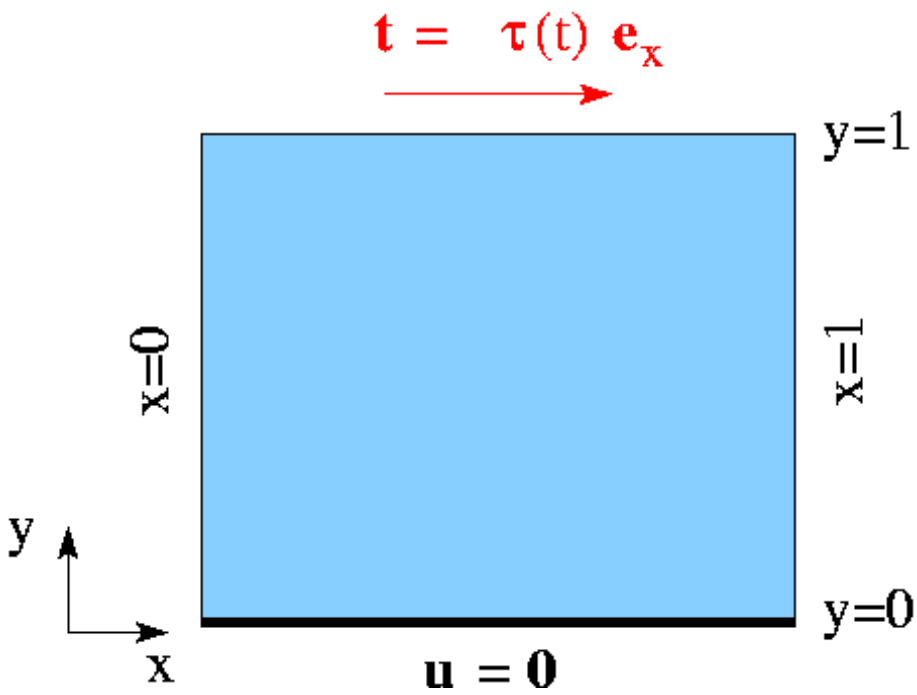

Figure 1.1 Sketch of the problem.

The flow is governed by the 2D unsteady Navier-Stokes equations,

$$Re\left(St\frac{\partial u_i}{\partial t}+u_j\frac{\partial u_i}{\partial x_j}\right)=-\frac{\partial p}{\partial x_i}+\frac{\partial}{\partial x_j}\left(\frac{\partial u_i}{\partial x_j}+\frac{\partial u_j}{\partial x_i}\right), \qquad (2)$$

and

$$\frac{\partial u_i}{\partial x_i}=0,$$

in the square domain

$$D=\Big\{(x_1,x_2)\mid x_1\in[0,1],\ x_2\in[0,1]\Big\}.$$

We apply the Dirichlet (no-slip) boundary condition

$$\mathbf{u}|_{\partial D_{lower}}=(0,0), \qquad (3)$$

on the lower, stationary wall, $\partial D_{lower}=\big\{(x_1,x_2)\mid x_2=0\big\}$ and the traction

$$\mathbf{t}^{[prescribed]}=(\tau(t),0) \qquad (4)$$

where $\tau(t)$ is a given function, on the upper boundary, $\partial D_{upper}=\big\{(x_1,x_2)\mid x_2=1\big\}$. As in the `previous example` we apply periodic boundary conditions on the "left" and "right" boundaries:

$$\mathbf{u}|_{x_1=0}=\mathbf{u}|_{x_1=1}\,. \qquad (5)$$

Initial conditions for the velocities are given by

$$\mathbf{u}(x_1,x_2,t=0)=\mathbf{u}_{IC}(x_1,x_2),$$

where $\mathbf{u}_{IC}(x_1,x_2)$ is given.

1.1.1 An exact, parallel-flow solution

We choose the prescribed traction $\mathbf{t}^{[prescribed]} = (\tau(t), 0)$ such that the parallel-flow solution

$$\mathbf{u}_{exact}(x_1, x_2, t) = U(x_2, t)\,\mathbf{e}_1 \quad \text{and} \quad p_{exact}(x_1, x_2, t) = 0,$$

derived in the `previous example` remains valid. For this purpose we set

$$\tau(t) = \left.\frac{\partial U(x_2, t)}{\partial x_2}\right|_{x_2=1},$$

where

$$U(x_2, t) = Re\left\{\frac{e^{i\omega t}}{e^{i\lambda} - e^{-i\lambda}}\left(e^{i\lambda y} - e^{-i\lambda y}\right)\right\}$$

and

$$\lambda = i\sqrt{i\omega Re\, St}.$$

1.2 Results

The two animations below show the computed solutions obtained from a spatial discretisation with Taylor-Hood and Crouzeix-Raviart elements, respectively. In both cases we set $\omega = 2\pi,\ Re = ReSt = 10$ and specified the exact, time-periodic solution as the initial condition, i.e. $\mathbf{u}_{IC}(x_1, x_2) = \mathbf{u}_{exact}(x_1, x_2, t = 0)$. The computed solutions agree extremely well with the exact solution throughout the simulation.

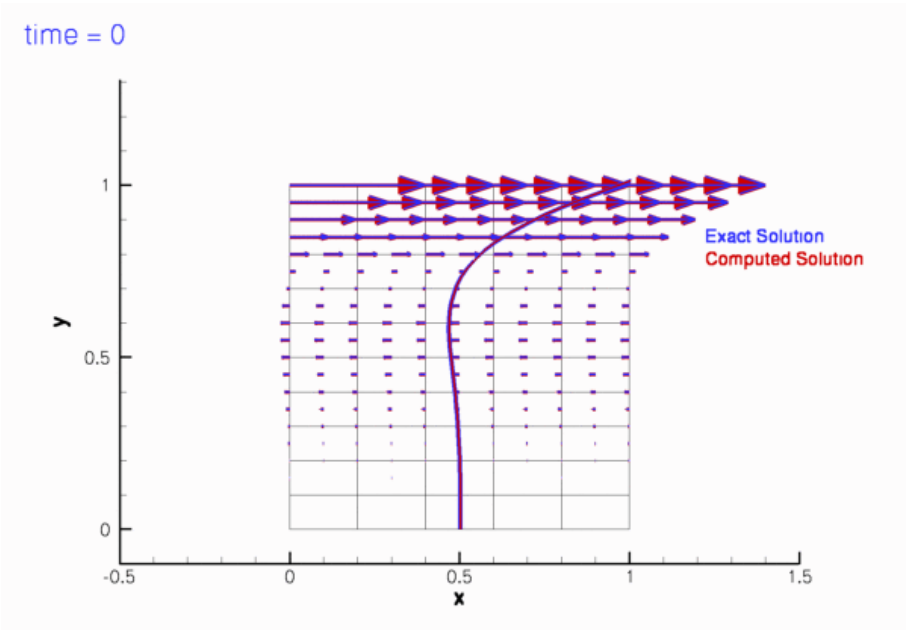

Figure 1.2 Plot of the velocity field computed with 2D Crouzeix-Raviart elements, starting from the exact, time-periodic solution.

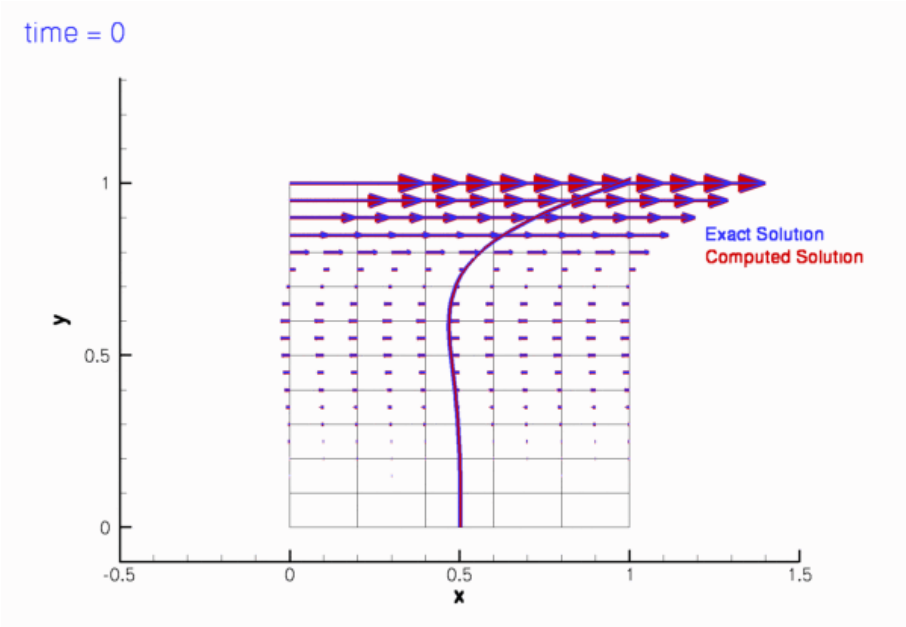

Figure 1.3 Plot of the velocity field computed with 2D Taylor-Hood elements, starting from the exact, time-periodic solution.

1.3 The global parameters

As usual, we use a namespace to define the problem parameters, the Reynolds number, Re, and the Womersley number, $Re\,St$. We also provide two flags that indicate the length of the run (to allow a short run to be performed when the code is run as a self-test), and the initial condition (allowing a start from $\mathbf{u}_{exact}$ or an impulsive start in which the fluid is initially at rest).

```
//===start_of_namespace=================================================
/// Namepspace for global parameters
//======================================================================
namespace Global_Parameters
{
 /// Reynolds number
 double Re;

 /// Womersley = Reynolds times Strouhal
 double ReSt;

 /// Flag for long/short run: Default =  perform long run
 unsigned Long_run_flag=1;

 /// Flag for impulsive start: Default = start from exact
 /// time-periodic solution.
 unsigned Impulsive_start_flag=0;

} // end of namespace
```

1.4 The exact solution

We use a second namespace to define the time-periodic, parallel flow $\mathbf{u}_{exact}$, and the traction $\mathbf{t}^{[prescribed]}$ required to make $\mathbf{u}_{exact}$ a solution of the problem.

```
//==start_of_exact_solution============================================
/// Namespace for exact solution
//======================================================================
namespace ExactSoln
{

 /// Exact solution of the problem as a vector
 void get_exact_u(const double& t, const Vector<double>& x, Vector<double>& u)
 {
  double y=x[1];
  // I=sqrt(-1)
  complex<double> I(0.0,1.0);
  // omega
  double omega=2.0*MathematicalConstants::Pi;
  // lambda
```

```
  complex<double> lambda(0.0,omega*Global_Parameters::ReSt);
  lambda = I*sqrt(lambda);

  // Solution
  complex<double> sol(
   exp(complex<double>(0.0,omega*t)) *
   (exp(lambda*complex<double>(0.0,y))-exp(lambda*complex<double>(0.0,-y)))
   /(exp(I*lambda)-exp(-I*lambda)) );

  // Extract real part and stick into vector
  u.resize(2);
  u[0]=real(sol);
  u[1]=0.0;

 }

 /// Traction required at the top boundary
 void prescribed_traction(const double& t,
                          const Vector<double>& x,
                          const Vector<double> &n,
                          Vector<double>& traction)
 {
  double y=x[1];
  // I=sqrt(-1)
  complex<double> I(0.0,1.0);
  // omega
  double omega=2.0*MathematicalConstants::Pi;
  // lambda
  complex<double> lambda(0.0,omega*Global_Parameters::ReSt);
  lambda = I*sqrt(lambda);

  // Solution
  complex<double> sol(
   exp(complex<double>(0.0,omega*t)) *
   (exp(lambda*complex<double>(0.0,y))+exp(lambda*complex<double>(0.0,-y)))
   *I*lambda/(exp(I*lambda)-exp(-I*lambda)) );

  //Extract real part and stick into vector
  traction.resize(2);
  traction[0]=real(sol);
  traction[1]=0.0;
 }

}  // end of exact_solution
```

1.5 The driver code

As in the previous example we use optional command line arguments to specify which mode the code is run in: Either as a short or a long run (indicated by the first command line argument being 0 or 1, respectively), and with initial conditions corresponding to an impulsive start or a start from the time-periodic exact solution (indicated by the second command line argument being 1 or 0, respectively). If no command line arguments are specified the code is run in the default mode, specified by parameter values assigned in the namespace Global_Parameters.

```
//===start_of_main=====================================================
/// Driver code for Rayleigh-type problem
//=====================================================================
int main(int argc, char* argv[])
{


 /// Convert command line arguments (if any) into flags:
 if (argc==1)
  {
   cout « "No command line arguments specified -- using defaults." « std::endl;
  }
 else if (argc==3)
  {
   cout « "Two command line arguments specified:" « std::endl;
   // Flag for long run
   Global_Parameters::Long_run_flag=atoi(argv[1]);
   /// Flag for impulsive start
   Global_Parameters::Impulsive_start_flag=atoi(argv[2]);
  }
 else
  {
   std::string error_message =
    "Wrong number of command line arguments. Specify none or two.\n";
   error_message +=
    "Arg1: Long_run_flag [0/1]\n";
   error_message +=
    "Arg2: Impulsive_start_flag [0/1]\n";

   throw OomphLibError(error_message,
                       OOMPH_CURRENT_FUNCTION,
```

```
                        OOMPH_EXCEPTION_LOCATION);
  }
 cout « "Long run flag: "
      «  Global_Parameters::Long_run_flag « std::endl;
 cout « "Impulsive start flag: "
      «  Global_Parameters::Impulsive_start_flag « std::endl;
```

Next we set the physical and mesh parameters.

```
 // Set physical parameters:

 // Womersly number = Reynolds number (St = 1)
 Global_Parameters::ReSt = 10.0;
 Global_Parameters::Re = Global_Parameters::ReSt;

 //Horizontal length of domain
 double lx = 1.0;

 //Vertical length of domain
 double ly = 1.0;

 // Number of elements in x-direction
 unsigned nx=5;

 // Number of elements in y-direction
 unsigned ny=10;
```

Finally we set up `DocInfo` objects and solve for both Taylor-Hood elements and Crouzeix-Raviart elements.

```
 // Solve with Crouzeix-Raviart elements
 {
  // Set up doc info
  DocInfo doc_info;
  doc_info.number()=0;
  doc_info.set_directory("RESLT_CR");

  //Set up problem
  RayleighTractionProblem<QCrouzeixRaviartElement<2>,BDF<2> >
   problem(nx,ny,lx,ly);

  // Run the unsteady simulation
  problem.unsteady_run(doc_info);
 }


 // Solve with Taylor-Hood elements
 {
  // Set up doc info
  DocInfo doc_info;
  doc_info.number()=0;
  doc_info.set_directory("RESLT_TH");

  //Set up problem
  RayleighTractionProblem<QTaylorHoodElement<2>,BDF<2> >
   problem(nx,ny,lx,ly);

  // Run the unsteady simulation
  problem.unsteady_run(doc_info);
 }

} // end of main
```

1.6 The problem class

The problem class remains similar to that in the `previous example`. Since we are no longer driving the flow by prescribing a time-periodic tangential velocity at the upper wall, the function `actions_before_implicit↩_timestep()` can remain empty.

```
//===start_of_problem_class=============================================
/// Rayleigh-type problem: 2D channel flow driven by traction bc
//======================================================================
template<class ELEMENT, class TIMESTEPPER>
class RayleighTractionProblem : public Problem
{
public:

 /// Constructor: Pass number of elements in x and y directions and
 /// lengths
 RayleighTractionProblem(const unsigned &nx, const unsigned &ny,
                 const double &lx, const double &ly);

 /// Update before solve is empty
 void actions_before_newton_solve() {}

 /// Update after solve is empty
```

```
 void actions_after_newton_solve() {}

 /// Actions before timestep (empty)
 void actions_before_implicit_timestep() {}

 /// Run an unsteady simulation
 void unsteady_run(DocInfo& doc_info);

 /// Doc the solution
 void doc_solution(DocInfo& doc_info);

 /// Set initial condition (incl previous timesteps) according
 /// to specified function.
 void set_initial_condition();
```

The function `create_traction_elements(...)` (discussed in more detail below) creates the traction elements and "attaches" them to the specified boundary of the "bulk" mesh.

```
 /// Create traction elements on boundary b of the Mesh pointed
 /// to by bulk_mesh_pt and add them to the Mesh object pointed to by
 /// surface_mesh_pt
 void create_traction_elements(const unsigned &b,
                               Mesh* const &bulk_mesh_pt,
                               Mesh* const &surface_mesh_pt);
```

The traction boundary condition sets the pressure so the function `fix_pressure(...)` used in the `previous example` is no longer required. The problem's private member data contains pointers to the bulk and surface meshes and the output stream that we use to record the time-trace of the solution.

```
private:

 /// Pointer to the "bulk" mesh
 RectangularQuadMesh<ELEMENT>* Bulk_mesh_pt;

 /// Pointer to the "surface" mesh
 Mesh* Surface_mesh_pt;

 /// Trace file
 ofstream Trace_file;

}; // end of problem_class
```

1.7 The problem constructor

We start by building the timestepper, determining its type from the class's second template argument, and pass a pointer to it to the Problem, using the function `Problem::add_time_stepper_pt(...)`.

```
//===start_of_constructor=============================================
/// Problem constructor
//====================================================================
template<class ELEMENT,class TIMESTEPPER>
RayleighTractionProblem<ELEMENT,TIMESTEPPER>::RayleighTractionProblem
(const unsigned &nx, const unsigned &ny,
 const double &lx, const double& ly)
{
 //Allocate the timestepper
 add_time_stepper_pt(new TIMESTEPPER);
```

Next we build the periodic bulk mesh,

```
 //Now create the mesh with periodic boundary conditions in x direction
 bool periodic_in_x=true;
 Bulk_mesh_pt =
  new RectangularQuadMesh<ELEMENT>(nx,ny,lx,ly,periodic_in_x,
                                   time_stepper_pt());
```

and the surface mesh,

```
 // Create "surface mesh" that will contain only the prescribed-traction
 // elements. The constructor just creates the mesh without
 // giving it any elements, nodes, etc.
 Surface_mesh_pt = new Mesh;
```

and use the function `create_traction_elements(...)` to populate it with traction elements that attach themselves to the specified boundary (2) of the bulk mesh.

```
 // Create prescribed-traction elements from all elements that are
 // adjacent to boundary 2, but add them to a separate mesh.
 create_traction_elements(2,Bulk_mesh_pt,Surface_mesh_pt);
```

We add both sub-meshes to the `Problem`, using the function `Problem::add_sub_mesh(...)` and use the function `Problem::build_global_mesh()` to combine the sub-meshes into the `Problem's` single, global mesh.

```
 // Add the two sub meshes to the problem
```

```
add_sub_mesh(Bulk_mesh_pt);
add_sub_mesh(Surface_mesh_pt);

// Combine all submeshes into a single Mesh
build_global_mesh();
```

We apply Dirichlet boundary conditions where required: No-slip on the stationary, lower wall, at $x_2 = 0$, parallel outflow on the left and right boundaries, at $x_1 = 0$ and $x_1 = 1$. No velocity boundary conditions are applied at the "upper" boundary, at $x_2 = 1$, where the traction boundary condition is applied.

```
// Set the boundary conditions for this problem: All nodes are
// free by default -- just pin the ones that have Dirichlet conditions
// here
unsigned num_bound=Bulk_mesh_pt->nboundary();
for(unsigned ibound=0;ibound<num_bound;ibound++)
 {
  unsigned num_nod=Bulk_mesh_pt->nboundary_node(ibound);
  for (unsigned inod=0;inod<num_nod;inod++)
   {
    // No slip on bottom
    // (DO NOT PIN TOP BOUNDARY, SINCE TRACTION DEFINES ITS VELOCITY!)
    if (ibound==0)
     {
      Bulk_mesh_pt->boundary_node_pt(ibound,inod)->pin(0);
      Bulk_mesh_pt->boundary_node_pt(ibound,inod)->pin(1);
     }
    // Horizontal outflow on the left (and right -- right bc not
    // strictly necessary because of symmetry)
    else if ((ibound==1) || (ibound==3))
     {
      Bulk_mesh_pt->boundary_node_pt(ibound,inod)->pin(1);
     }
   }
 } // end loop over boundaries
```

Next we pass the pointers to the Reynolds and Strouhal numbers, Re, $Re\,St$, to the bulk elements.

```
//Complete the problem setup to make the elements fully functional
//Loop over the elements
unsigned n_el = Bulk_mesh_pt->nelement();
for(unsigned e=0;e<n_el;e++)
 {
  //Cast to a fluid element
  ELEMENT *el_pt = dynamic_cast<ELEMENT*>(Bulk_mesh_pt->element_pt(e));

  //Set the Reynolds number, etc
  el_pt->re_pt() = &Global_Parameters::Re;
  el_pt->re_st_pt() = &Global_Parameters::ReSt;
 }
```

Finally we pass pointers to the applied traction function to the traction elements and assign the equation numbers.

```
// Loop over the flux elements to pass pointer to prescribed traction function
// and pointer to global time object
n_el=Surface_mesh_pt->nelement();
for(unsigned e=0;e<n_el;e++)
 {
  // Upcast from GeneralisedElement to traction element
  NavierStokesTractionElement<ELEMENT> *el_pt =
   dynamic_cast< NavierStokesTractionElement<ELEMENT>*>(
    Surface_mesh_pt->element_pt(e));

  // Set the pointer to the prescribed traction function
  el_pt->traction_fct_pt() =
   &ExactSoln::prescribed_traction;
 }

//Assgn equation numbers
cout « assign_eqn_numbers() « std::endl;

} // end of constructor
```

1.8 Create traction elements

The creation of the traction elements is performed exactly as in the `Poisson` and `unsteady heat` problems with flux boundary conditions, discussed earlier. We obtain pointers to the "bulk" elements that are adjacent to the specified boundary of the bulk mesh from the function `Mesh::boundary_element_pt(...)`, determine which of the elements' local coordinate(s) are constant along that boundary, and pass these parameters to the constructors of the traction elements which "attach" themselves to the appropriate face of the "bulk" element. Finally, we store the pointers to the newly created traction elements in the surface mesh.

```
//============start_of_create_traction_elements=========================
/// Create Navier-Stokes traction elements on the b-th boundary of the
/// Mesh object pointed to by bulk_mesh_pt and add the elements to the
```

```
/// Mesh object pointeed to by surface_mesh_pt.
//======================================================================
template<class ELEMENT,class TIMESTEPPER>
void RayleighTractionProblem<ELEMENT,TIMESTEPPER>::
create_traction_elements(const unsigned &b, Mesh* const &bulk_mesh_pt,
                         Mesh* const &surface_mesh_pt)
{
 // How many bulk elements are adjacent to boundary b?
 unsigned n_element = bulk_mesh_pt->nboundary_element(b);

 // Loop over the bulk elements adjacent to boundary b?
 for(unsigned e=0;e<n_element;e++)
  {
   // Get pointer to the bulk element that is adjacent to boundary b
   ELEMENT* bulk_elem_pt = dynamic_cast<ELEMENT*>(
    bulk_mesh_pt->boundary_element_pt(b,e));

   //What is the index of the face of element e along boundary b
   int face_index = bulk_mesh_pt->face_index_at_boundary(b,e);

   // Build the corresponding prescribed-flux element
   NavierStokesTractionElement<ELEMENT>* flux_element_pt = new
    NavierStokesTractionElement<ELEMENT>(bulk_elem_pt,face_index);

   //Add the prescribed-flux element to the surface mesh
   surface_mesh_pt->add_element_pt(flux_element_pt);

  } //end of loop over bulk elements adjacent to boundary b

} // end of create_traction_elements
```

1.9 Initial conditions

The function `set_initial_conditions(...)` remains the same as in the `previous example`.

1.10 Post processing

The function `doc_solution(...)` remains the same as in the `previous example`.

1.11 The timestepping loop

The function `unsteady_run(...)` remains the same as in the `previous example`, except that the default number of timesteps is increased to 500.

1.12 Comments and Exercises

1.12.1 How do the traction elements work?

The finite element solution of the Navier-Stokes equations is based on their weak form, obtained by weighting the stress-divergence form of the momentum equations

$$Re\left(St\frac{\partial u_i}{\partial t}+u_j\frac{\partial u_i}{\partial x_j}\right)=\frac{\partial \tau_{ij}}{\partial x_j}, \qquad (6)$$

with the global test functions ψ_l, and integrating by parts to obtain the discrete residuals

$$f_{il}=\int_D\left[Re\left(St\frac{\partial u_i}{\partial t}+u_j\frac{\partial u_i}{\partial x_j}\right)\,\psi_l+\tau_{ij}\,\frac{\partial \psi_l}{\partial x_j}\right]dV-\int_{\partial D}\tau_{ij}\,n_j\,\psi_l\,dS=0. \qquad (7)$$

The volume integral in this residual is computed by the "bulk" Navier-Stokes elements. `Recall` that in the residual for the i -th momentum equation, the global test functions ψ_l vanish on those parts of the domain boundary ∂D where the i -th velocity component is prescribed by Dirichlet boundary conditions. On such boundaries, the surface integral in (7) vanishes because $\psi_l = 0$. If the velocity on a certain part of the domain boundary, $\partial D_{natural} \subset \partial D$, is not prescribed by Dirichlet boundary conditions and the surface integral over $\partial D_{natural}$ is not added to the discrete residual, the velocity degrees of freedom on those boundaries are regarded as unknowns and the "traction-free" (or natural) boundary condition

$$\tau_{ij}\,n_j=0 \quad \text{on} \quad \partial D_{natural}$$

is "implied". Finally, traction boundary conditions of the form (1) may be applied along a part, $\partial D_{traction} \subset \partial D$, of the domain boundary. The surface integral along this part of the domain boundary is given by

$$\int_{\partial D_{traction}} \tau_{ij}\, n_j\, \psi_l\, dS = \int_{\partial D_{traction}} t_i^{[prescribed]}\, \psi_l\, dS, \qquad (8)$$

where $t_i^{[prescribed]}$ is given, and it is this contribution that the traction elements add to the residual of the momentum equations.

1.12.2 Exercises

1. Pin the vertical velocity along the upper boundary, $x_2 = 1$, and compare the results against those obtained with the original version of the code. How does the change affect the velocity field? Why is pressure likely to change?
2. Pin the horizontal velocity along the upper boundary, $x_2 = 1$, and start the simulation with an impulsive start. Compare the results against those obtained with the original version of the code and explain your findings, referring to the theory provided in the section How do the traction elements work? .
3. Run the code with an impulsive start and confirm that it takes longer for the solution to approach the time-periodic solution than in the previous case where the flow was driven by the wall motion. [Here are some animations of the velocity fields obtained following an impulsive start, for a discretisation with Taylor-Hood elements and Crouzeix-Raviart elements.]
4. Investigate the effects of applying a non-zero value for t_2 on the top boundary.

1.13 Source files for this tutorial

- The source files for this tutorial are located in the directory:

  demo_drivers/navier_stokes/rayleigh_traction_channel/

- The driver code is:

  demo_drivers/navier_stokes/rayleigh_traction_channel/rayleigh_↩ traction_channel.cc

1.14 PDF file

A pdf version of this document is available. \